

ANAEMIA

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Reviewed in May 2024 by Dr Justin Chong. Next review date: May 2027.

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DEFINITION

Haemoglobin (Hb) Levels to Diagnose Anaemia at Sea Level

Population	Anaemia (g/dL)		
	Mild	Moderate	Severe
Children 6 - 59 months of age	10.0 - 10.9	7.0 - 9.9	< 7.0
Children 5 - 11 years of age	11.0 - 11.4	8.0 - 10.9	< 8.0
Children 12 - 14 years of age	11.0 - 11.9	8.0 - 10.9	< 8.0
Non-pregnant women (15 years of age and above)	11.0 - 11.9	8.0 - 10.9	< 8.0
Pregnant women	10.0 - 10.9	7.0 - 9.9	< 7.0
Men (15 years of age and above)	11.0 - 12.9	8.0 - 10.9	< 8.0

(Adapted from Reference 1, WHO 2011)

CLASSIFICATION

1. Classification of Anaemia Based on Causes

A) Production

- Haemoglobin synthesis disturbances: Iron deficiency, thalassaemia, anaemia of chronic disease, sickle cell disease
- Stem cell disease: Aplastic anaemia, myeloproliferative disorder, myelodysplastic syndrome, sideroblastic anaemia
- DNA synthesis disturbances: Folate deficiency, Vitamin B₁₂ deficiency, immunosuppressive agent
- Bone marrow infiltration: leukaemia, lymphoma, secondary malignancy, myelofibrosis
- Bone marrow suppression: Alcoholism, chemotherapy, hypothyroidism, radiotherapy
- Lack of erythropoietin: Chronic renal failure

B) Destruction

- Intrinsic haemolysis [due to red blood cell (RBC) Defects]: Spherocytosis, sickle cell disease, enzyme deficiencies
- Extrinsic haemolysis: Infection, autoimmune haemolytic anaemia, drugs, toxins, mechanical valves, haemolytic-uraemic syndrome, hypersplenism, haemangioma, transfusion reaction

C) Blood Loss

- Excessive menstruation
- Bleeding in the gastrointestinal or genitourinary tracts
- Trauma

D) Chronic Diseases

- Organ failure: Chronic renal failure, chronic liver disease
- Acute infections: Infective mononucleosis, viral hepatitis, Parvovirus infection, cytomegalovirus infection
- Chronic infections: Tuberculosis, human immunodeficiency virus infection, infective endocarditis
- Chronic inflammatory disorders: Rheumatoid arthritis, collagen vascular disease, polymyalgia rheumatica
- Malignancy
- Protein energy malnutrition

2. Classification of Anaemia Based on RBC Cell Size and Appearance

A) Microcytic Anaemia (MCV < 80 fl)

- Iron deficiency (50% presents as HCMC Anaemia)
- Thalassaemia
- Anaemia of chronic disease
- Sideroblastic anaemia
- Lead poisoning

B) Macrocytic Anaemia (MCV > 100 fl)

- Megaloblastic Anaemia:
 - Vitamin B₁₂ deficiency
 - Folate deficiency
 - Drugs:
 - Anticonvulsants, e.g., Phenytoin, Phenobarbitone
 - Antibiotics, e.g., Bactrim, Oral contraceptives
 - Others – Metformin, Methotrexate
- Non-megaloblastic Anaemia:
 - Alcoholism
 - Hypothyroidism
 - Chronic liver disease
 - Melodysplastic syndrome
 - Reticulocytosis
 - Malignancy with marrow infiltration

C) Normocytic Anaemia (80-100 fl)

- Anaemia of inflammation
- Mixed iron and Vitamin B₁₂ ± folate deficiency
- Hypothyroidism
- Leukaemia
- Aplastic Anaemia
- Haemolytic Anaemia
- Haemorrhages
- Hypersplenism
- RBC membrane defect
- Autoimmune destruction

CLINICAL ASSESSMENT OF ANAEMIA

Clinical Presentation

- Asymptomatic
- Fatigue
- Light headedness
- Headache
- Cognitive impairment
- Chest pain
- Palpitation
- Breathlessness
- Numbness or coldness of extremities

Physical Examination & Look Out for Related Causes

- Pallor of conjunctivae and lips, palmar creases
- Jaundice (haemolytic anaemia or chronic liver disease or alcoholism)
- Angular cheilosis, glossitis, koilonychias (iron deficiency)
- Thalassaemia facies (frontal bossing, chipmunk facies)
- Goitre (hypothyroidism)
- Compensatory tachycardia, low blood pressure
- Parasternal systolic ejection flow murmur
- Stigmata of chronic liver disease (palmar erythema, hepatic asterixis, spider naevi)
- Organomegaly (myeloproliferative disorder, infiltrative disease, polycystic kidneys, space occupying lesions)
- Renal causes (sallow features, uremic breath, AV fistula, transplanted kidney)
- Per rectal examination (bleeding, haemorrhoids, rectal masses)
- Skin (petechiae rashes, bruises, ecchymosis)
- Lymphadenopathy (lymphoproliferative disorder)
- Autoimmune conditions (SLE, RA, scleroderma)

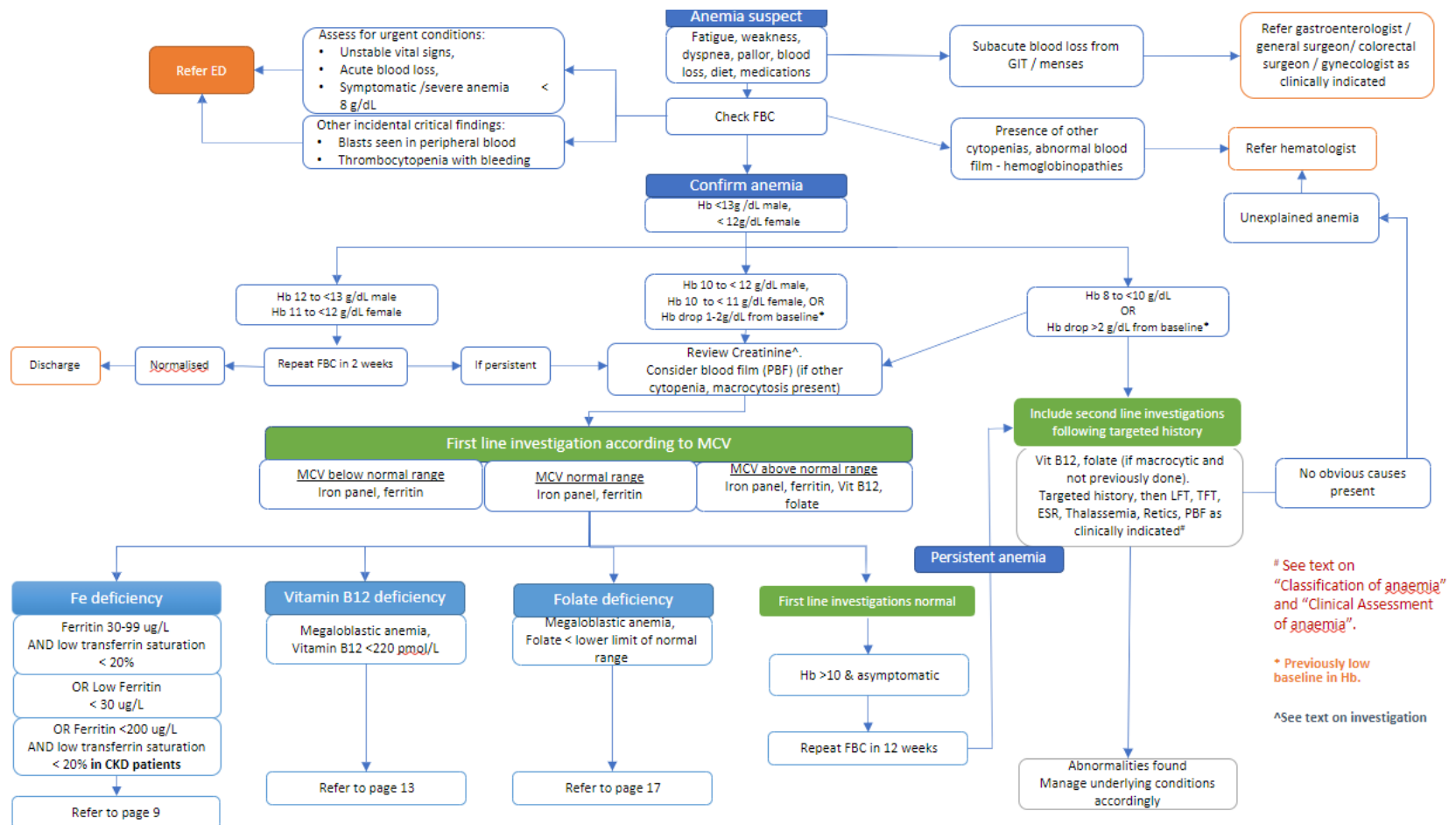
Investigations

- If clinically suspect anaemia, to do full blood count (FBC).
- If Hb < 12 g/dL in males or < 11 g/dL in female, for further evaluation base on the level of presenting haemoglobin. (Refer to flow chart on [Approach to Anaemia](#))
- A drop in Hb by >1 g/dL from previously low baseline is significant and should be further evaluated.
- Patients with borderline low Hb at onset, to trend the Hb in 2 weeks. If normalised, no further workup required.
- First line investigations include iron panel, ferritin, and in the presence of macrocytosis, include vitamin B₁₂ and folate. Review latest creatinine for presence of Chronic Kidney Disease (CKD), looking back up to one year if there is no reason to expect worsening renal function. In the absence of haematological disorders or deficiencies causing the anaemia, along with Hb <10g/dL and CKD G3/G4/G5, routine nephrology appointment should be considered ([NUP CKD CPG](#)).
- Peripheral blood film (PBF) is useful in the presence of cytopenias or macrocytosis to check for megaloblastic anaemia or myelodysplastic syndrome. It is recommended when there is macrocytosis and other cytopenias, OR macrocytosis with Hb <10g/dL. In the presence of macrocytosis with normal / slightly low Hb without an overt cause, a PBF is helpful as well.
- If no obvious cause can be found, screen clinically for other causes of anaemia – thyroid disease, liver disease, malignancy, personal or family history of thalassaemia, and chronic inflammatory diseases.
- In the absence of causes found from history, physical exam and 1st line investigations, and a stable patient with isolated mild anaemia who is asymptomatic, a repeat FBC can be arranged in 3 months' time with return or emergency advice provided.
- Second line investigations may be guided by targeted history, and include retic count, thyroid function test, liver function test, thalassaemia screen, ESR.
- A high retic count of > 3% (in the absence of recent haematinic replenishment for haematinic deficiency) implies possible haemolytic anaemia which requires referral to the haematologist.

General Management

- Acute haemorrhage, symptomatic or severe anaemia
 - Rapid resuscitation with crystalloid, refer Emergency Department after patient is haemodynamically stable.
- Suspected haematologic disease or unexplained anaemia after investigations:
 - Refer to haematologist.
- Manage nutritional deficiencies if present
 - Refer to section on [Specific Management](#).

APPROACH FOR ANAEMIA



SPECIFIC MANAGEMENT

Iron Deficiency Anaemia

- Iron is found in the heme group of haemoglobin that carries oxygen, that supports respiration, energy production, DNA synthesis and cell proliferation.
- Deficiency in iron is usually chronic and asymptomatic. Before iron deficiency is severe enough to cause microcytic anaemia, patients may have non-specific symptoms like weakness, fatigue, difficulty concentrating or poor work productivity. However, there is no strong evidence for replacement therapy in cases of iron deficiency without anaemia.

Causes

Systems	Causes
Physiologic	Infancy, puberty, menstruation, pregnancy, blood donation
Diet	Vegetarian, insufficient intake from poverty, malnutrition.
Gastrointestinal (Blood loss / reduced absorption)	Upper GI: Helicobacter pylori infection, stomach surgery, atrophic gastritis, Crohn disease, gastritis, peptic ulcer disease, cancer Lower GI: Diverticulitis, benign tumour, GI cancer, inflammatory bowel disease, (Crohn's, ulcerative colitis), angiodysplasia, haemorrhoids, hookworm infestation.
Gynaecological /genitourinary	Heavy menses, haematuria, haemoglobinuria
Medications	Salicylates, NSAIDS, proton-pump inhibitors, glucocorticoids.
Genetic	Iron-refractory iron-deficiency anaemia.
Iron-restricted erythropoiesis	Treatment with erythropoiesis-stimulating agents, anaemia of chronic disease, chronic kidney disease

Interpretation of iron studies

- Serum ferritin level is the most efficient and cost-effective test to gauge iron store.

Point of care	Iron studies interpretation in patient with no advance renal failure* or inflammatory disease			
	Ferritin level, µg/L	Transferrin Saturation	Interpretation	Management
At presentation	< 30	< 20%	Iron deficiency with no iron store	Start oral iron replacement
	30-99	< 20%	Iron deficiency with some iron store	Start oral iron replacement
	≥ 100	≥ 20%	Fe deficiency unlikely	Evaluate for other causes
Monitoring of iron therapy	< 100	< 20%	Iron deficiency present	Continue oral replacement
	30-99	≥ 20%	Recent oral iron replacement initiation	Continue oral iron replacement
	≥ 100	< 20%	Suspect functional iron deficiency	Evaluate for anaemia of chronic disease
	≥ 100	≥ 20%	Fe deficiency resolved	If anaemia resolves, stop therapy and discharge. If anaemia persists, to evaluate for other causes.

*Note: In patients with renal failure (CKD G3a and worse), iron deficiency is defined as ferritin < 200 µg/L and transferrin saturation < 20%.

- Iron studies interpretation and management in patients with anaemia and CKD G3a and worse

	Ferritin level, µg/L	Transferrin Saturation	Interpretation	Management
At presentation	<200	< 20%	Iron deficiency with low / no iron store	Start oral iron replacement
	≥ 200	< 20%	Functional iron deficiency	Start oral iron replacement
	≥ 200	≥ 20%	Fe deficiency unlikely Suspect anaemia of chronic kidney disease	If Hb ≥ 10, evaluate further and KIV refer Haem If Hb < 10, refer Renal for management KIV erythropoietin stimulating agents (ESA)
Monitoring of iron therapy	< 200	< 20%	Iron deficiency present	Continue oral replacement. If no response of Hb after 3-6 months of oral replacement, refer to Haem or Renal.*
	< 200	≥ 20%	Recent oral iron replacement initiation	Continue oral iron replacement. If no response of Hb after 3-6 months of oral replacement, refer to Haem or Renal.*
	≥ 200	< 20%	Suspect functional iron deficiency	Refer to Renal if anaemia persists. For KIV ESA and/or IV iron if ferritin < 300.
	≥ 200	≥ 20%	Fe deficiency resolved	If anaemia resolves, stop therapy and discharge. If anaemia persists, refer to Renal. Kiv ESA.

Notes:

- 1) ESA – erythropoiesis stimulating agent, e.g., Recormon, Darbepoietin, Mircera. ESA and IV iron are generally initiated at Renal or Haem clinic.
- 2) CKD patients with iron deficiency should be checked for underlying causes as per Iron Deficiency flowchart.
- 3) * Refer to Haem if no other indication for Renal referral; refer Renal if there are other indications for Renal referral.

Oral iron therapy

- Recommended daily oral iron replacement dosing:
 - 100 - 200 mg of elementary iron, or
 - 3 mg/kg per day as liquid iron preparation to infants and children. (Liquid iron preparation is not available in NUP)
- Start at low dose once a day to avoid initial intolerance, preferably on empty stomach with monitoring of side effects. Ferrous gluconate and iron polymaltose formulations may be taken during or after meals, as recommended by the manufacturers.
- Side effects include nausea, vomiting, constipation, metallic taste, dark greenish stools (does not affect stool occult blood test).
- For those who are intolerant to iron tablet, strategies to reduce adverse effect include:
 - Take the tablets after meals.
 - Reduce the dose of the iron tablets. Alternative-day dosing is acceptable too.
 - Use a stool softener or bulk-forming laxative.
- Strategies to improve iron absorption include:
 - Take separately from foods that may impair absorption, e.g., coffee, tea, milk, and dairy products.
 - Minimise exposure to medications that reduce gastric acidity, e.g., proton pump inhibitors and histamine receptor blockers. Iron should be given 2 hours before, or 4 hours after, ingestion of antacids.
 - Combine with vitamin C.

Table 1. Oral iron replacement therapy

<u>Iron salt</u>	<u>Elemental iron, mg</u>	<u>Starting dose</u>	<u>Availability at NUP</u>
Ferrous gluconate	30	1 tablet BD	√
Iron polymaltose	100	1 tablet OD	√

Response to treatment

- A response is seen in 7-10 days with a rise in haemoglobin (Hb) level by at least 1 g/dL
- 3 months of treatment can be required for **normalisation of Hb**.
- Up to 6 months of treatment can be required for **repletion of iron stores**. It is thus recommended to continue iron therapy for another 3 months after Hb level normalises to replenish iron store.

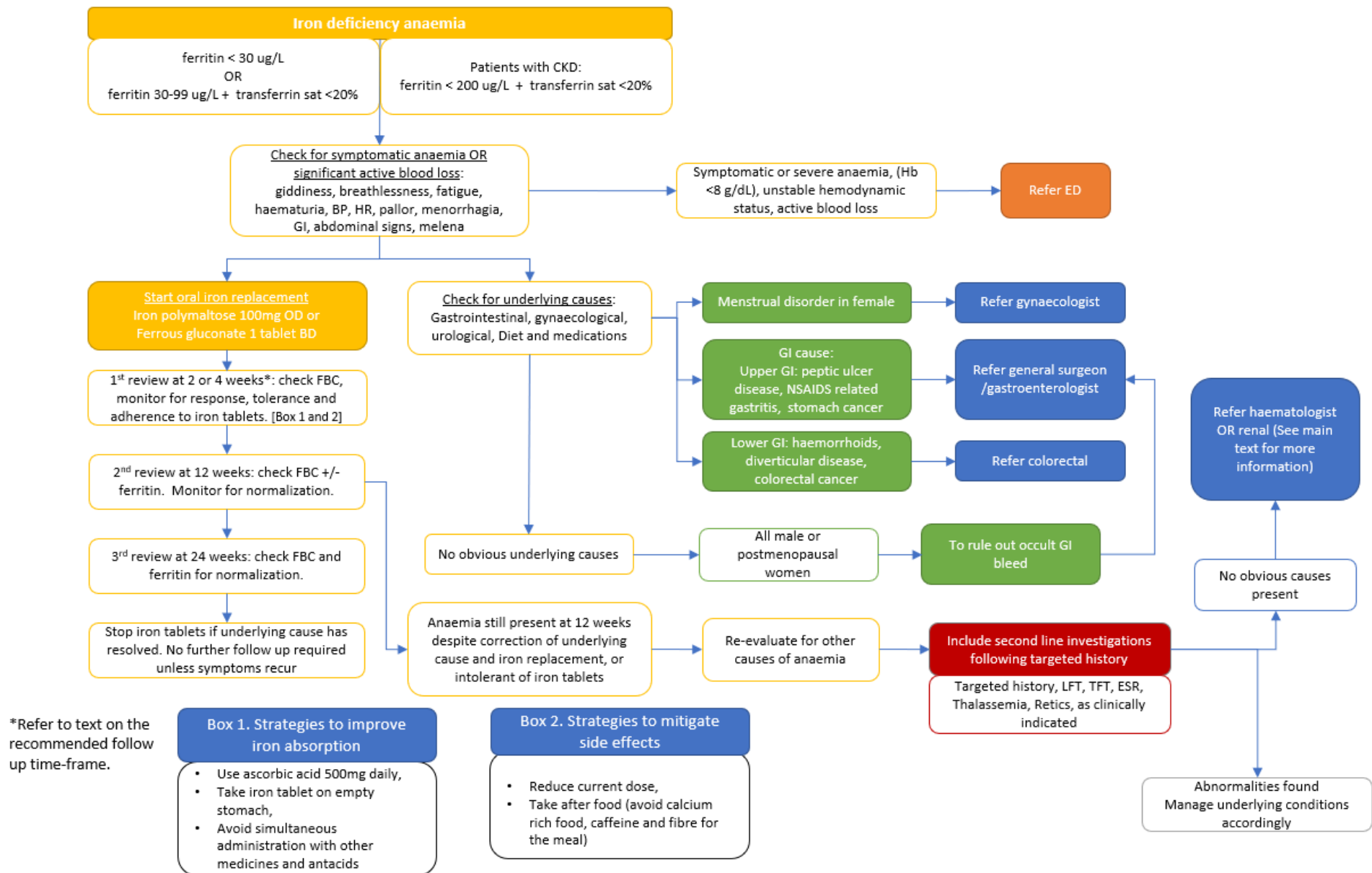
Monitoring and follow-up

- First follow up with FBC at 4 weeks to check for rise in Hb, adherence and side effects to iron therapy and review underlying causes. Advise for early review at 2 weeks if low Hb of 8 to 10 g/dL, no improvement in symptoms or patient is intolerant to iron therapy.
- If response is adequate, follow up with FBC ($\pm$ serum ferritin) can be at 12 weeks, focusing on normalisation of Hb. A further follow up at 24 weeks with FBC and serum ferritin for normalisation in ferritin level and ensuring Hb remains normal.
- Stop iron replacement after Hb and ferritin are normalised, and cause of iron loss is resolved.
- No further monitoring is needed unless symptoms recur.
- A lack of response to oral iron replacement can be due to the following reasons:
 - Premature termination of treatment, either by the physician or patient
 - Lack of adherence to the iron tablets (to check for side effects)
 - Unresolved underlying cause of iron deficiency
 - True refractory response to treatment

Further evaluation and referral

- Refer for further evaluation and management of underlying causes. The possible referrals are as follows:
 - Dietitian: dietary causes.
 - Gynaecologist: abnormal menstrual bleeding.
 - Gastroenterologist / general surgeon: patients with gastrointestinal related symptoms, risk factors and comorbidities.
 - Colorectal surgeon: for patients with per rectal bleeding.
- For all men and post-menopausal women with no obvious cause for iron deficiency anaemia, a referral to gastroenterologist or general surgery is indicated for endoscopy to rule out occult gastrointestinal blood loss.

Approach to Iron Deficiency Anaemia



Vitamin B₁₂ Deficiency

- Vitamin B₁₂ (Vit B₁₂) deficiency develops over years due to large body stores. The daily allowance for adults is 2.4 mcg and can be found in fish, meats, and fortified cereals.
- Vit B₁₂ plays a role in red blood cell production. About 10% of vitamin B₁₂ deficiency manifest as anaemia.
- Vit B₁₂ deficiency can also manifest as neuropsychological symptoms alone without anaemia due to its role in DNA synthesis, the myelination process, and the maintenance of the central nervous system.

Common clinical presentation

System	Clinical Presentation
Neuropsychiatric	Impaired cognition, mood disorders, symmetrical paraesthesia/ numbness, gait problems.
Haematological	Macrocytic anaemia (commonly), normochromic anaemia (if iron deficiency coexists), pancytopenia (in severe cases) Hypersegmented neutrophils, megaloblastic feature on blood film
Glossopharyngeal	Glossitis, taste impairment

Risk profiles of patients with higher risk of Vitamin B₁₂ deficiency

- Restricted diet
 - Long term vegetarians, elderly, poor dentition, alcohol abuse.
- Prolonged use of medications that can decrease Vit B₁₂ absorption
 - metformin > 4month, proton pump inhibitor / histamine H₂ blocker use > 12 months
- Decreased intrinsic factor
 - Pernicious anaemia, atrophic gastritis, post gastrectomy syndrome.
- Decrease ileal absorption
 - Crohn's disease, ileal resection.

Vitamin B₁₂ levels

Vitamin B ₁₂ levels, pmol/L	Interpretation	Management
< 145	Low. Deficiency present (97% sensitive)	Start replacement
145 - 220	Borderline. Vitamin B ₁₂ deficiency cannot be ruled out in the presence of symptoms	Start replacement if symptoms or anaemia present
220- 569	Normal	Evaluate for other causes

Replacement therapy of Vitamin B₁₂ deficiency

- Injectable therapy leads to more rapid improvement and should be considered in patients with severe deficiency, severe neurological symptoms, malabsorption, and compliance issues. Replacement treatment is with 1000 mcg intramuscular injections of cyanocobalamin given - every day or every other day for 1 week, followed by once-weekly for 4-8 weeks.
- Patients who present with neurological symptoms require step down parenteral injection to fortnightly for another 6 months after replacement therapy.
- Consider consultation with a gastroenterologist if planning to convert to oral supplementation.
- Those with reversible causes should be treated until the deficiency is corrected and symptoms resolve. Those with long term causes of vitamin B₁₂ deficiency should be switched to maintenance therapy once symptoms / anaemia resolve.
- If patient has both folate and vitamin B₁₂ deficiencies, vitamin B₁₂ should be replaced first to prevent subacute combined degeneration of the spinal cord.

Treatment response and monitoring

- The clinical response to therapy is usually marked – an immediate sense of well-being, rapid reversal of pancytopenia and mucosal changes, bone marrow reverting to normal in 24 hours, and a brisk reticulocytosis can be seen as early as after 3rd day of treatment.
- Patients with neurological symptoms or other cytopenias should be reviewed at 1 week with FBC.
 - At 1 week, there should be haemoglobin increase by $\geq 1\text{g/dL}$, as well as resolution of leukopenia and thrombocytopenia if present.
 - If there is no improvement in anaemia, leukopenia, or thrombocytopenia by 1 week, refer patient to the haematologist.
- FBC may be repeated at 4 weeks, and 8 weeks if not normalised at 4 weeks from the start of treatment to assess for response.
 - If anaemia has not normalised by 8 weeks, refer patient to the haematologist.

- Neurological recovery begins within 1 week and complete response can be expected within 6 to 12 weeks of treatment. A small percentage may have residual disability, especially if diagnosis and treatment are delayed by 6 months or more.
- After treatment is stopped, it is recommended to monitor Vitamin B₁₂ and FBC in one year's time to ensure levels are within normal.
- Following which, the general guide for monitoring of B₁₂ levels:
 - Restricted diet which is reversible: No further monitoring of B₁₂ required unless clinically indicated (symptomatic anaemia, neuropsychological symptoms etc.)
 - On chronic medications that can cause mild malabsorption (i.e., metformin): 2-yearly B₁₂ monitoring
 - Vegetarian, known pernicious anaemia, OR chronic conditions that can cause severe malabsorption: Yearly FBC and B₁₂ monitoring

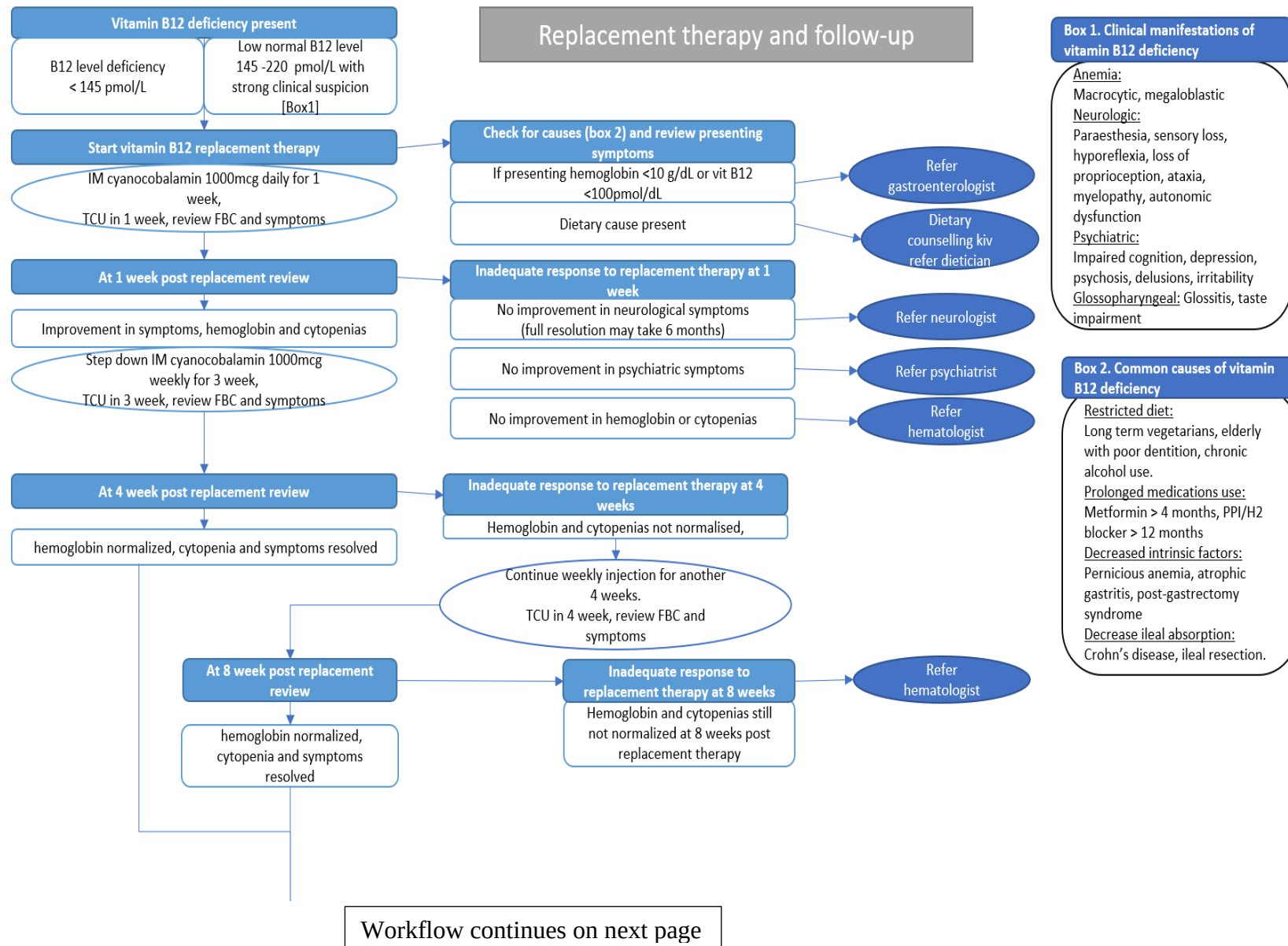
Management of underlying causes

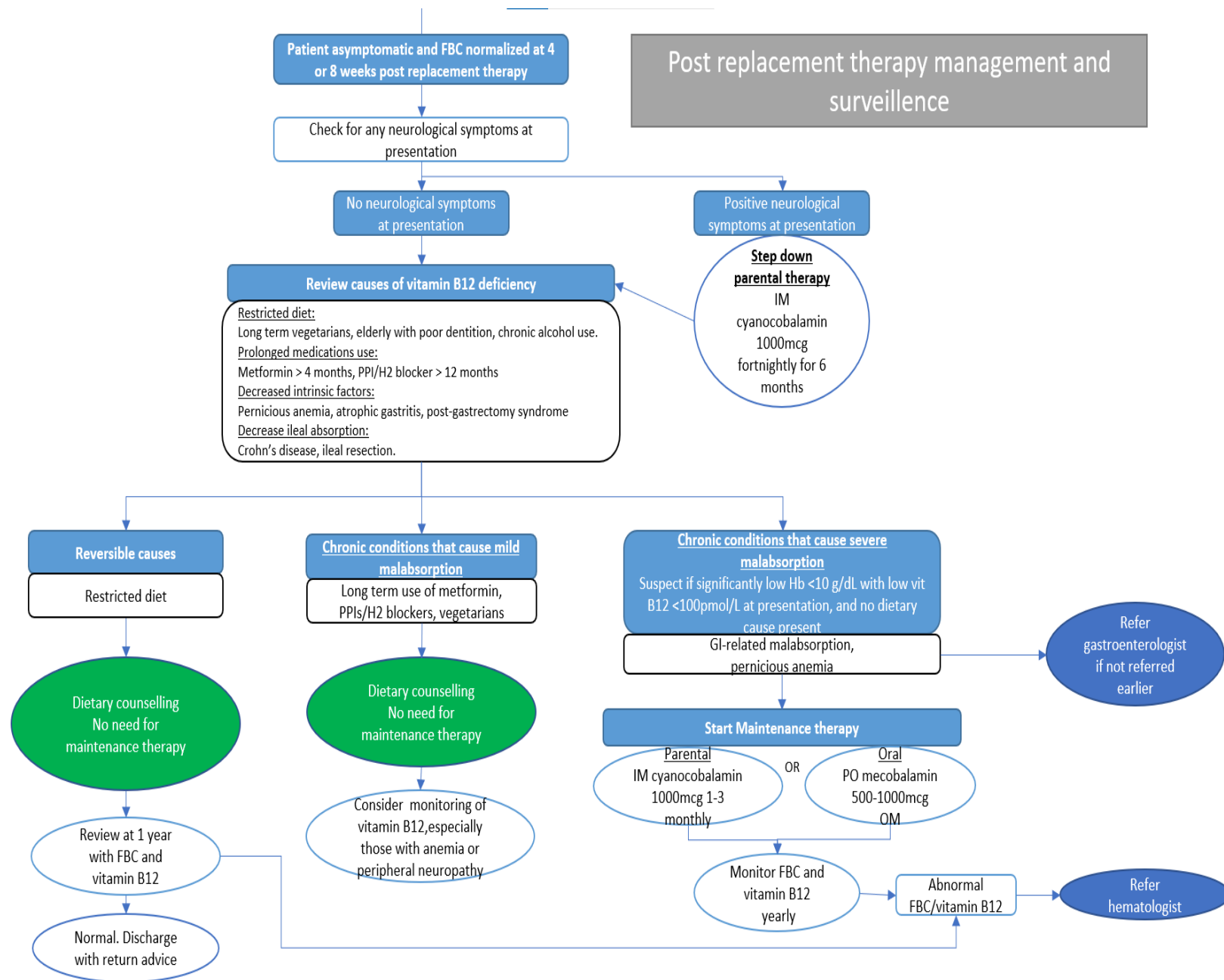
- Refer to dietitian for patients with dietary causes.
- Refer to the relevant specialities for patients with neuropsychiatric symptoms.
- Refer to haematologist if cytopenias and/or haemoglobin have not normalised by 8 weeks.
- Refer patients with gastrointestinal-related malabsorption, suspected pernicious anaemia or inflammatory bowel disorder to gastroenterologist for further evaluation. Features suggestive of pernicious anaemia include severe vitamin B₁₂ deficiency anaemia at presentation with no obvious dietary or malabsorptive aetiology, does not respond well to replacement therapy, or has coexisting autoimmune conditions.

Lifelong maintenance therapy of Vitamin B₁₂ deficiency

- Lifelong maintenance treatment is required for patients with severe malabsorption conditions like chronic gastrointestinal related malabsorption, and pernicious Anaemia.
- Usual maintenance is with 1000 mcg injections at 1 to 3-monthly intervals.
- Daily oral supplementation with Vitamin B₁₂ 1000 mcg daily can be as effective as IM injections.
- Monitor FBC and serum B₁₂ yearly while on maintenance therapy.
- Doses can be titrated according to B₁₂ levels, FBC and symptoms.
 - If B₁₂ ≥ 800 pmol/L, withhold B₁₂ replacement and monitor with FBC and vitamin B₁₂ in 6 months.
 - In the absence of pernicious anaemia, vitamin B₁₂ stores in the body takes up to 2 years to deplete.
 - Patients with pernicious anaemia may deplete faster at around 6 months.
 - Restart maintenance therapy at lower dose (i.e., half the frequency) when vitamin B₁₂ < 800 pmol/L, or if recurrence of macrocytic anaemia / symptoms.

- Resume annual FBC and B₁₂ monitoring, or earlier based on clinical judgement.
- After replacement therapy, those with mild malabsorption issue due to metformin or PPI usage, or long-term vegetarians can be advised on dietary intake of vitamin B₁₂ and to be monitored annually for symptoms of anaemia or numbness. Laboratory monitoring to follow the general guide described in the section on “treatment response and monitoring” if no clinical suspicion of B₁₂ deficiency.





Folate Deficiency

- Folate is found in plant-based foods and fortified grains.
- Folate deficiency develops quicker over weeks to months due to limited body stores and high turnover rate during normal cell division.
- Folate deficiency causes megaloblastic anaemia and impaired cognition. Although neuropsychiatric changes have been reported in folate deficiency, such symptoms are more commonly described in vitamin B₁₂ deficiency. Additionally, lack of folate during early pregnancy increases the risk of neural tube defects.

Causes

Mechanism	Conditions
Nutritional deficiency	Substance abuse, overcooked foods, alcoholism, poor quality diet, goats' milk
Malabsorption	Celiac disease, inflammatory bowel disease, GI malignancy, short bowel syndrome
Drugs	Methotrexate, trimethoprim, phenytoin, sulphasalazine
Increased requirements	Pregnancy, lactation, chronic haemolysis, exfoliative dermatitis

Folate level

Folate level, nmol/L	Interpretation	Action
Within normal laboratory reference range	Normal	Evaluate for other causes
Less than lower limit of normal laboratory reference range	Low	Start folic acid replacement

Treatment and follow-up

- Rule out presence of vitamin B₁₂ deficiency before starting therapy.
- Initial treatment – 5 mg daily until correction of anaemia (usually by 8 weeks).
- All patients should be investigated for causes of folate deficiency.

- FBC may be repeated at 4 week, and 8 weeks (if haemoglobin remains low at 4 weeks) from the start of treatment to assess for response.
- At 1 week, there should be increase in retic count and haemoglobin and resolution of leukopenia and thrombocytopenia if present.
- If leukopenia / thrombocytopenia or anaemia has not normalised by 8 weeks, refer patient to the haematologist.
- Stop folate therapy after 8 weeks of replacement or until anaemia normalised, whichever is later. No further monitoring of FBC or serum folate is required.
- Only patients with chronic haemolysis or increased cell turnover disorders need long-term therapy. This group of patients is usually followed up at their respective specialists. Otherwise, monitoring with FBC and folate yearly is recommended.

Anaemia of Chronic Disease (ACD)

Features

- Anaemia with normal or low MCV
- Presence of a chronic inflammatory condition, e.g., infection, autoimmune disease, kidney disease or cancer.
- Evidence of functional iron deficiency: low to normal iron, low transferrin, low transferrin saturation; ferritin $\geq 100\mu\text{g/L}$.
- Other causes of anaemia excluded.

Risk profiles of patients with higher risk of anaemia of chronic disease

- Collagen vascular and autoimmune disorders:
 - Rheumatoid arthritis, systemic lupus erythematosus (SLE), inflammatory bowel disease
- Chronic infections:
 - Tuberculosis, chronic fungal infection, hepatitis, HIV
- Chronic disease:
 - Moderate or severe renal disease, diabetes mellitus, congestive heart failure, chronic pulmonary disease, chronic liver disease, cerebral vascular disease
- Cancer:
 - Leukaemia, lymphoma, renal cell carcinoma, multiple myeloma, solid tumour
- Critical illness and major trauma

Clinical assessment

- Past medical history
 - Chronic diseases or recent major surgery / trauma / critical illness.
- System review
 - Joint pain and swelling, fever and constitutional symptoms of anorexia, night sweats, arthralgia, myalgia, weight loss.
- Physical examination
 - Check for lymph nodes, abdominal tenderness, and mass, organomegaly, chest examination, rash, joints, and eye.
- First line Investigation: (done as part of anaemia workup)
 - FBC, ferritin, iron studies, vitamin B₁₂ / folate, review creatinine, PBF if macrocytosis or other cytopenias present.
- Second line investigation:
 - KIV ESR / creatinine, retics, LFT, TFT if clinically indicated.

Treatment approach

- Treat underlying cause. Refer to relevant specialist if required.
- Refer to the haematologist if the diagnosis is unclear or in established cases of ACD when Hb < 10 g/dL and the underlying cause cannot be eliminated. Further treatment may involve erythropoietin +/- parenteral iron.

INDICATIONS FOR REFERRAL TO HAEMATOLOGIST IN PATIENTS WITH ANAEMIA

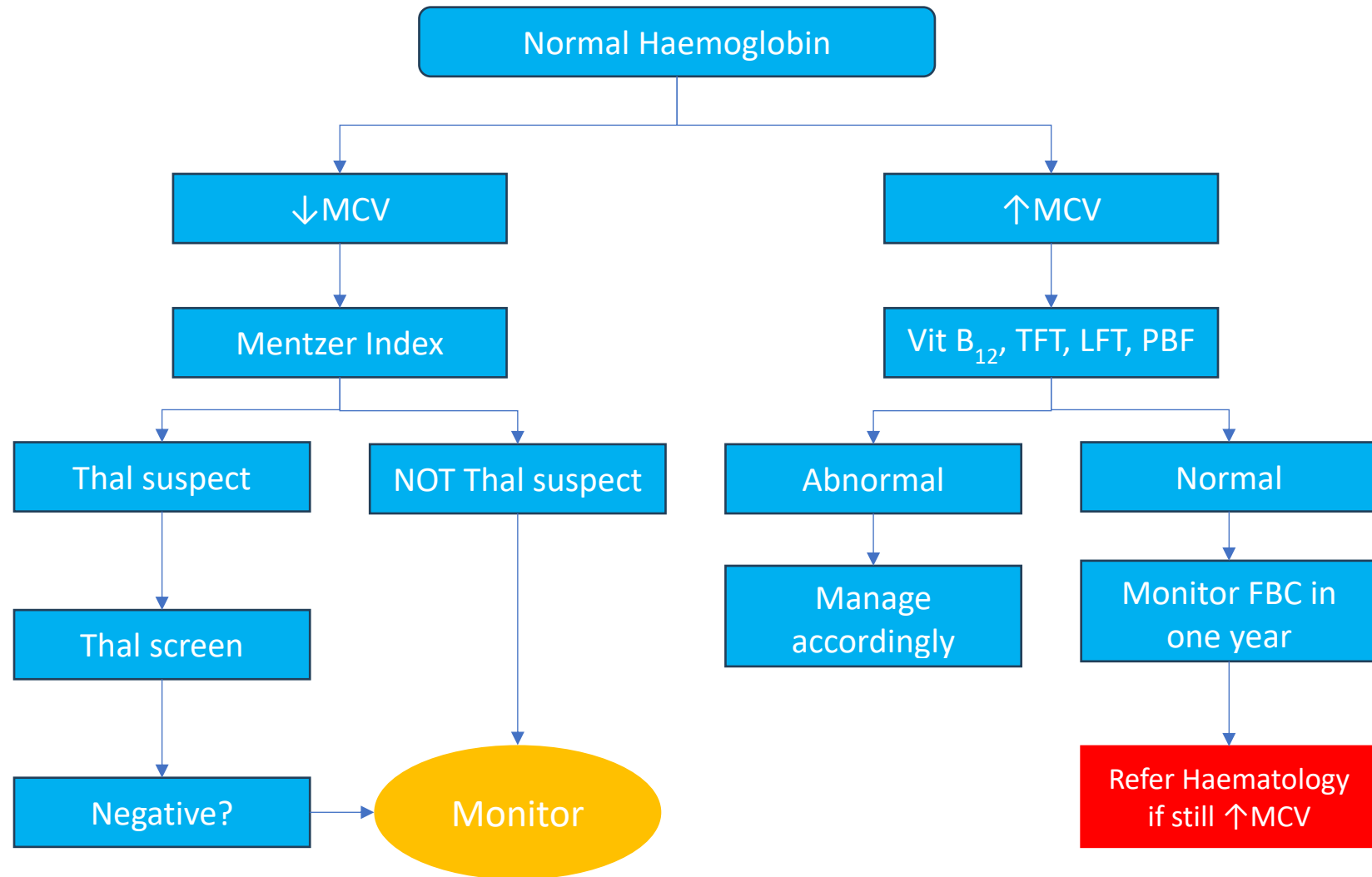
Point of care	Indications for referral
At presentation	• Haematological emergencies (via ED): symptomatic anaemia, severe anaemia, blast, thrombocytopenia with bleeding. • Presence of other cytopenias [myelodysplastic syndrome], abnormal blood film – haemoglobinopathies (except thalassaemia).
After investigations	• Unexplained anaemia
Failure to respond to replacement therapy	• Persistent Fe deficiency anaemia at 3 months and no other causes of anaemia found. • Persistent Vit B₁₂ deficiency related anaemia with other cytopenias at 1 week. • Persistent Vit B₁₂ / folate deficiency anaemia at 8 weeks [macrocytic anaemia – to rule out marrow disease]
Long-term follow-up	• Relapse of vit B₁₂ deficiency anaemia after stopping of therapy [to rule out pernicious anaemia if not previously done] • Hb < 10 in those with anaemia of chronic disease or CKD
Other indications	• Persistent macrocytosis without anaemia (1 year follow up) • Retic > 3% (suspect hemolytic anaemia)

ANNEXES: QUICK REFERENCE GUIDE

Annex A: Other Haematological-Related Issues

Haematological issues	Possible causes	Clues	Actions
Normal haemoglobin with microcytosis	Thalassaemia trait	Mentzer index (MCV/ RBC) < 13, RDW < 14% normal	Consider Thalassaemia workup
Normal haemoglobin with macrocytosis	Normal variant	Stable MCV with no progression	Exclude other causes at onset.
	Early B ₁₂ deficiency	blood film for hypersegmented neutrophils, may be normal	Empirical therapy with neurobion in high-risk patients, repeat FBC in 6 months, no monitoring if normalised. Otherwise refer haematologist
	Early myelodysplastic syndrome, Subclinical thyroid disease, liver disease	Assess clinically Check blood film, KIV TFT and LFT	Manage causes clinically. If no known cause, monitor FBC in 1 year. - If persistent macrocytosis, refer haematologist. - If normalised, no further investigations required
High ferritin while on iron replacement	<u>Mild – mod Iron overload</u> No chronic inflammation/CKD: Female > 150 µg/L, male > 400 µg/L Chronic inflammation/CKD: > 500 µg/L	Transferrin saturation > 50%, patient on iron supplements	Stop iron supplement Repeat ferritin in 3 - 6 months Refer haematologist if ferritin remains high after repeat
	<u>Severe iron overload</u> > 1000 µg/L With or without chronic inflammation/ CKD	Transferrin saturation > 50%	Stop iron supplement Refer haematologist for further evaluation.

Approach to Normal Hemoglobin Level with Abnormal MCV



Annex B: Iron, B₁₂, Folate Replacement Regimes

	Updates to NUP CPG – Fe Deficiency			
Fe deficiency cut-off	Ferritin level < 30 OR Ferritin level 30 - 99 + transferrin sat < 20% OR CKD patients: Ferritin level < 200 + transferrin sat < 20%			
Replacement	Options:			
	Iron salt	Elemental iron, mg	Starting dose	Availability at NUP
	Ferrous gluconate	30	1 tablet BD	✓
	Iron polymaltose	100	1 tablet OD	✓
	(Please refer to the section on Oral Iron Therapy of NUP CPG)			
Monitoring	Time frame for monitoring after starting replacement: <u>At 2* or 4 weeks:</u> FBC – rise in Hb, symptoms improvement (advice to return earlier within 2 weeks if still symptomatic or intolerant of Fe tablet) (*earlier TCU for patients with Hb of 8 to 10 g/dL) <u>At 12 weeks:</u> FBC ± ferritin – normalisation of Hb → <i>if Hb still low, refer haematologist</i> <u>At 24 weeks:</u> FBC + ferritin → Normal Hb and ferritin: stop Rx → Low Hb or low ferritin: continue replacement and refer haematologist			
Follow-up after therapy	If underlying causes resolved, discharge with return advice.			

	Updates to NUP CPG – Vit B ₁₂ deficiency
Vitamin B12 cutoff	B₁₂ deficient: • Lab cutoff: < 145, OR • < 220 pmol/L (in the presence of symptoms)
Replacement	Acute: - Parenteral therapy (refer to main text for indication): 1000mcg i/m* cyanocobalamin given - every day or every other day for 1 week, followed by once-weekly for 4-8 weeks. (*deep subcutaneous injection instead, if thrombocytopenic) Patients with neurological symptoms require step down parenteral injection to fortnightly for another 6 months after replacement therapy. - Oral therapy (refer to main text for indication): 1000mcg oral mecobalamin OD for 6 months Maintenance: 1000mcg i/m cyanocobalamin given 1 to 3 monthly, OR 1000mcg oral mecobalamin OD

	Updates to NUP CPG – Vit B ₁₂ deficiency
Monitoring while on replacement therapy	- NCIS step down care for vit B₁₂ for patients with megaloblastic anaemia and pernicious anaemia, and started on i/m B₁₂ treatment: Monitor with FBC at 4 weeks or 8 weeks. - If i/m B₁₂ treatment is initiated at polyclinic: <u>At 1 week:</u> FBC: Hb increase by ≥ 1g/dL + resolution of other cytopenias if present. Refer to haematologist if above is not achieved. <u>At 4 weeks:</u> FBC: Monitor Hb If still low, continue i/m B₁₂ treatment <u>At 8 weeks:</u> FBC: Monitor Hb and for any cytopenias If Hb still low or any cytopenias, refer to haematologist
Maintenance Rx	Monitor FBC and vitamin B₁₂ yearly if on maintenance therapy
Follow-up after stopping Rx	For patients with NO long-term causes of B12 deficiency and treatment can be stopped: Review with FBC + B₁₂ in 1 year. If normal, discharge with return advice.

	Updates to NUP CPG – Folate Deficiency
Folate deficiency cut-off	Lab cut-off: Less than lower limit of normal lab reference range
Replacement	Rule out presence of vitamin B ₁₂ deficiency before starting therapy Initial treatment – 5 mg daily until correction of anaemia (usually by 8 weeks)
Monitoring while on Rx	- 1 week review if other cytopenia and neuro symptom at presentation. - FBC at 4 weeks, check Hb, WBC and platelets. Refer Haematology if ↓WBC or ↓platelets. - FBC at 8 weeks if Hb was low at 4 weeks. Refer Haematology if still ↓Hb - If FBC normalises at 8 weeks, continue folate replacement for total of 3 to 6 months. - Discontinue folate thereafter if the patient does not have chronic haemolysis or increased cell turnover disorders. - Advise appropriate diet. - No further lab test needed.
Follow-up	If patient needs long term replacement and is not followed up specialist, yearly monitoring with FBC and folate.

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